THE CARBON CLASSIFICATION THEOREM: AN EXACT CLOSED-FORM CRITERION FOR WHEN A FLEXIBLE LOAD REDUCES EMISSIONS

TEAM A1 — ROLE 1 (OPTIMAL CONTRACT AND GENERAL CARBON THEORY)

Abstract. This is the Role 1 Week 4 deliverable: an exact, computable criterion for the carbon sign. We construct a carbon functional $\Phi(F)$—an explicit affine combination of the distribution-free cutoffs, the equilibrium capacities, the surplus probability $q^\circ = F(0; K_R^\circ)$, and exactly two conditional means of the residual-demand law—and show $\Delta = \Phi(F)$ *on the large-FCL regime* $K_M^* \geq K_R^*$, so there carbon falls if and only if $\Phi(F) > 0$. The regime hypothesis is essential, not cosmetic: outside it the integration by parts picks up a $K_M^* F(-K_M^*)$ term and Φ can carry the *wrong sign*—at $C_R = 0.34$ the closed form reports $+1.905$ while the true Δ is -0.135 (Remark 1). The identity is Week-2 bookkeeping; the value is that Φ is a *closed form on a named, minimal set of coordinates*, and that the sign needs distributional information no coarser than those coordinates: both conditional means enter with nonzero weight, and (Week 2) the sign is not decidable from capacities and cutoffs alone over the admissible-law class. We are explicit about scope. Φ is the criterion for the *naive* dispatch of Weeks 1–2, for which—by the team's own results—$\Phi < 0$ at interior optima (carbon rises) and $\Phi > 0$ only at the cheap-renewable corner. The point of naming Φ is exactly to *explain the Week-2→Week-3 flip*: the surplus contract of Week 3 zeroes the utilization term and turns Φ positive, so the classification is the ledger on which "why the optimal contract decarbonizes" is read off. The three questions that a scan alone had settled are now *conditional theorems*, each gated on one explicit, checkable condition that we then verify: (a) each mean is individually sign-decisive *iff* a scalar margin beats the carbon deficit $(\mathrm{D}_\ell, \mathrm{D}_m)$—an exact both-directions criterion; and (b) surplus-only is globally carbon-optimal ($\Lambda \leq 1$) under a transparent capacity-monotonicity condition (the budget raises fossil, not renewable, capacity—Condition M), which we *prove* forces $\Lambda \leq 0$. A third claim—that $\{\Phi = 0\}$ is a carbon boundary and the carbon-down set an interval under a monotonicity Condition S—is **withdrawn** (Remark 3): every cell in which the supporting sweep reported $\Phi > 0$ lies in the small-FCL regime where (3) is not the correct identity, and the true Δ is negative in all of them. Consequently the "only if" branch of the classification is never exercised on the present evidence: across every cell we can solve, $\Delta < 0$. Verification in `verify_week4.py` now carries a regime guard and raises rather than returning a wrong-signed Φ.

1. Setup: the affine representation

We keep the model and notation of Weeks 1–3. A regulated utility chooses renewable K_R, fossil K_F, and flexible-computing-load (FCL) K_M capacity; residual demand is $\tilde{x} = \tilde{D} - K_R\tilde{S} \in [-K_R, \bar{D}]$ with CDF $F(x; K_R)$, density $f > 0$ on the interior, $F_{K_R} := \partial F/\partial K_R > 0$. Assumption 1 ($R > c + C_M$, $R < c + C_F + C_M$, $\beta + p > c + C_F$, $\beta + p > R$) holds throughout. The distribution-free cutoffs are $\varphi_1 = F(a; K_R^*) = C_M/(R - c)$, $\varphi_2 = F(K_F^*; K_R^*) = (\beta + p - c - C_F - C_M)/(\beta + p - R)$, and $\varphi^\circ = F(K_F^\circ; K_R^\circ) = (\beta + p - c - C_F)/(\beta + p - c)$, with $a := K_F^* - K_M^*$.

The keystone of Week 2 is that net fossil is *affine* in two conditional means. Recall (Week 2, eq. (fos)) the exact identities

$$\mathbb{E}[\text{fossil}^*] = K_F^* - \tfrac{C_M}{R-c}\big(a - \mu_{\text{low}}^*\big), \qquad \mathbb{E}[\text{fossil}^\circ] = K_F^\circ(1 - \varphi^\circ) + (\varphi^\circ - q^\circ)\,\mu_{\text{mid}}^\circ, \tag{1}$$

where $q^\circ = F(0; K_R^\circ)$ and

$$\mu_{\text{low}}^* := \mathbb{E}[\tilde{x} \mid \tilde{x} \in (-K_R^*, a)] \in (-K_R^*, a), \qquad \mu_{\text{mid}}^\circ := \mathbb{E}[\tilde{x} \mid \tilde{x} \in (0, K_F^\circ)] \in (0, K_F^\circ) \tag{2}$$

Date: Working results, Week 4.

are the conditional means over the FCL surplus zone and the no-FCL mid zone. The net carbon saving is $\Delta := \mathbb{E}[\text{fossil}^\circ] - \mathbb{E}[\text{fossil}^*]$; carbon falls iff $\Delta > 0$.

2. The carbon classification theorem

Definition (The carbon functional)**.** At the equilibrium define

$$\Phi(F) \;:=\; \underbrace{\tfrac{C_M}{R-c}\big(a-\mu^*_{\text{low}}\big)}_{\text{capacity substitution}} \;-\; \underbrace{\Big[K^*_F - K^\circ_F(1-\varphi^\circ) - (\varphi^\circ - q^\circ)\,\mu^\circ_{\text{mid}}\Big]}_{\text{net utilization}}. \tag{3}$$

Φ is a closed-form affine functional of the statistic $T(F) := (K^*_F, K^*_M, K^*_R, K^\circ_F, \varphi^\circ, q^\circ, \mu^*_{\text{low}}, \mu^\circ_{\text{mid}})$: distribution-free cutoffs, the equilibrium capacities, and the two conditional means (2).

Theorem 1 (Carbon classification, large-FCL regime)**.** *Let F have an interior optimum* in the large-FCL regime*, $K^*_M \geq K^*_R$—equivalently $F(-K^*_M; K^*_R) = 0$. Then*

$$\boxed{\Delta = \Phi(F), \qquad \textit{hence} \qquad \textit{carbon falls} \iff \Phi(F) > 0.}$$

The equivalence is exact on that regime and holds in both directions.

Remark 1 (The hypothesis is not cosmetic: outside it, Φ can carry the wrong sign). Φ as defined in (3) is the large-FCL identity: the lower limit $-K^*_R$ and the normalizer $F(a)$ are correct only when $F(-K^*_M) = 0$. In small-FCL the integration by parts of (1) picks up an extra $K^*_M F(-K^*_M)$ term, and the discrepancy is $O(F(-K^*_M))$—large enough to flip the sign. Solving the equilibrium across C_R and comparing the closed form against direct quadrature:

C_R	regime	$F(-K^*_M)$	Φ	Δ	$\lvert\Phi - \Delta\rvert$
0.34	small-FCL	0.2706	+1.905	−0.135	2.04
0.36	small-FCL	0.1872	+0.280	−0.577	0.857
0.40	small-FCL	0.0832	−0.707	−0.950	0.242
0.46	small-FCL	0.0057	−1.546	−1.556	0.0096
0.52	large-FCL	0.0000	−2.358	−2.358	2.7×10^{-8}

The error tracks $F(-K^*_M)$ exactly, and at the two smallest C_R the closed form reports carbon *falling* when it is in fact *rising*. Any use of Φ must therefore carry the regime hypothesis; `verify_week4.py` now raises rather than returning a wrong-signed value.

Proof. Subtract the two identities in (1):

$$\Delta = \mathbb{E}[\text{fossil}^\circ] - \mathbb{E}[\text{fossil}^*] = K^\circ_F(1-\varphi^\circ) + (\varphi^\circ - q^\circ)\mu^\circ_{\text{mid}} - K^*_F + \tfrac{C_M}{R-c}(a - \mu^*_{\text{low}}),$$

which is exactly $\Phi(F)$ of (3). As $\Delta = \Phi$ identically, every sign statement transfers. Both directions are therefore immediate: $\Delta > 0 \Rightarrow \Phi > 0$ and $\Phi > 0 \Rightarrow \Delta > 0$.[1] □

The content of Theorem 1 is not the identity—which is Week-2 bookkeeping—but that the carbon sign is a *closed form on a named, small set of coordinates*: the distribution-free cutoff φ°, four capacities $K^*_F, K^*_M, K^*_R, K^\circ_F$ (equivalently the cutoffs φ_1, φ_2 and φ°), the surplus probability $q^\circ = F(0; K^\circ_R)$, and the two conditional means. Of these only φ° (and φ_1, φ_2) is distribution-free; the capacities, q°, and the means are all functionals of F—so Φ depends on F through both the equilibrium map and the two moments, and "coordinates" below means these entries, not a probabilistic sufficient statistic. Whether the sign reduces further—to the distribution-free cutoffs alone—is the subject of the next section, and the answer is no.

[1] Verified in `verify_week4.py`: the closed-form Φ of (3) equals the direct quadrature of $\mathbb{E}[\text{fossil}^\circ] - \mathbb{E}[\text{fossil}^*]$ to $\sim 10^{-9}$ at every optimum across the distribution library, and $\operatorname{sign}\Delta = \operatorname{sign}\Phi$ in every cell.

Corollary 1 (The sufficient conditions, recovered)**.** *The Week-1/2 sufficient condition follows by bounding the means:* $\mu^*_{\text{low}} \le (a - K^*_R)/2$ *and* $\mu^\circ_{\text{mid}} \ge K^\circ_F/2$ *(monotone-density shape class* $\mathcal{U}$*) give* $\Phi \ge \Phi_0 := \frac{C_M}{R-c}\frac{a+K^*_R}{2} - K^*_F + K^\circ_F[(1-\varphi^\circ) + \frac{\varphi^\circ - q^\circ}{2}]$*, so* $\Phi_0 > 0$ *implies carbon falls. The classification subsumes it: the sufficient condition is the midpoint slice of the exact criterion* $\Phi > 0$*.*

3. Minimality: the sign needs both means

Week 2 proved the carbon sign is not a function of the distribution-free cutoffs alone: over the class of admissible conditional laws sharing the equilibrium cutoffs and capacities, Δ ranges over an interval straddling zero. We record what this says about Φ's arguments.

Proposition 1 (Both means are load-bearing)**.** *As a function of* $(\mu^*_{\text{low}}, \mu^\circ_{\text{mid}})$*,*

$$\frac{\partial \Phi}{\partial \mu^*_{\text{low}}} = -\frac{C_M}{R-c} < 0, \qquad \frac{\partial \Phi}{\partial \mu^\circ_{\text{mid}}} = \varphi^\circ - q^\circ > 0,$$

both nonzero: Φ *depends nontrivially on each conditional mean, so neither can be dropped from the criterion. Consequently the distribution-free cutoffs and the capacities do* not *determine* $\operatorname{sign}\Delta$*—the sign is genuinely distributional, resolved only once the two means are known.*

Proof. The partials are read off (3); $\varphi^\circ - q^\circ > 0$ because $\varphi^\circ = F(K^\circ_F) > F(0) = q^\circ$ $(K^\circ_F > 0)$. Nonzero partials give the nontrivial dependence. That capacities and cutoffs alone leave the sign undetermined is Week-2's under-determination result over the admissible-conditional-law class (Prop. irreducible); we adopt its caveat verbatim—the extremal laws attaining $\Delta_{\min}, \Delta_{\max}$ are degenerate and need not be realized by an equilibrium F, so the claim is under-determination over that class, not that an equilibrium attains the extremes.[2] □

The per-mean question—does each mean *alone* flip the sign?—has a clean exact answer, conditional on a single scalar margin.

Theorem 2 (Per-mean decisiveness, sharp)**.** *At an interior optimum, where* $\Delta = \Phi < 0$ *(Week-2),* μ^*_{low} *is individually sign-decisive—sweeping it over* $(-K^*_R, a)$ *with* μ°_{mid} *and the cutoffs held at equilibrium makes* $\operatorname{sign}\Phi$ *flip*—if and only if

$$\frac{C_M}{R-c}\,(\mu^*_{\text{low}} + K^*_R) \;>\; |\Delta| \tag{D_ℓ}$$

(the margin by which raising μ^*_{low} *to the support edge can lift* Φ *exceeds the deficit* $|\Delta|$*). Symmetrically,* μ°_{mid} *is individually sign-decisive iff* $(\varphi^\circ - q^\circ)(K^\circ_F - \mu^\circ_{\text{mid}}) > |\Delta|$ (D_m)*. When* (D_ℓ) *and* (D_m) *both hold, neither mean is redundant and the coordinate set of* Φ *is minimal.*

Proof. Φ is affine and strictly decreasing in μ^*_{low} (slope $-C_M/(R-c)$), so over $\mu^*_{\text{low}} \in (-K^*_R, a)$ it ranges over $(\Phi_{\text{lo}}, \Phi_{\text{hi}})$ with $\Phi_{\text{hi}} = \Delta + \frac{C_M}{R-c}(\mu^*_{\text{low}} + K^*_R)$ (at the upper edge $\mu^*_{\text{low}} = -K^*_R$) and $\Phi_{\text{lo}} = \Delta - \frac{C_M}{R-c}(a - \mu^*_{\text{low}}) < \Delta < 0$ (at the lower edge). The interval contains 0 iff $\Phi_{\text{hi}} > 0$, i.e. iff (D_ℓ). The μ°_{mid} statement is identical with slope $\varphi^\circ - q^\circ > 0$ and edge $\mu^\circ_{\text{mid}} = K^\circ_F$. Under both, each mean's sweep crosses zero, so a statistic dropping either cannot decide $\operatorname{sign}\Delta$.[3] □

Remark 2 (What is now pinned, and what remains). The per-mean flip is now a *theorem*, exact in both directions, gated on the explicit scalar margins $(\mathrm{D}_\ell), (\mathrm{D}_m)$—no longer a bare numerical observation. What is not proved is that $(\mathrm{D}_\ell), (\mathrm{D}_m)$ hold at *every* interior optimum (they hold

[2] Verified in `verify_week4.py`: at the representative optimum, sweeping μ^*_{low} alone (with μ°_{mid} and the cutoffs at their equilibrium values) makes Φ cross zero—$\Phi \in [-3.3, +0.4]$—and sweeping μ°_{mid} alone makes it cross—$\Phi \in [-5.3, +0.5]$; so at this optimum each mean is individually sign-decisive. Whether the single-mean interval contains zero at *every* interior optimum we do not prove.

[3] Verified in `verify_week4.py`: at the representative optimum (D_ℓ) and (D_m) hold with margins $\Phi_{\text{hi}} = +0.40$ and $+0.49$, so both means are individually decisive there.

wherever the sweep margin beats the carbon deficit, which is checkable per instance and holds throughout the scan). This is the honest boundary: the criterion's minimality is characterized, and its one ungoverned input is a transparent inequality on equilibrium quantities.

4. The carbon phase diagram

Remark 3 (Retracted: the carbon boundary was an artifact of the regime). An earlier draft stated a phase-diagram theorem: that $\{\Phi = 0\}$ is the exact carbon boundary and, under a monotonicity hypothesis (Condition S), the carbon-down set is an interval $\{C_R < C_R^{\dagger}\}$; the supporting evidence was a sweep reporting Φ falling monotonically from $+1.9$ to -2.5 with a single sign change.

That result is withdrawn. By Remark 1, every cell in which the sweep reported $\Phi > 0$ lies in the *small*-FCL regime, where (3) is not the correct identity; the true Δ computed by quadrature is *negative* in all of them. There is therefore no observed carbon-down region on this family, and $\{\Phi = 0\}$ separates nothing: the apparent boundary was the point at which a large-FCL formula, applied out of regime, changed sign.

Two consequences. First, Condition S is vacuous as stated, since the monotone crossing it described does not exist once Φ is evaluated only where it is valid. Second, the classification's "if and only if" cannot be exercised on this evidence: across every cell we can solve, $\Delta < 0$, so only the "carbon rises" branch is ever tested. We state the identity of Theorem 1 on its regime and make no claim about a boundary until a genuine carbon-improving equilibrium is exhibited.

5. Closing the frontier leverage condition

Week 3 reduced global carbon-optimality of the surplus contract over the budget family $d_\tau(x) = \min\{(-x)_+ + \tau, K_M, (K_F - x)_+\}$ to the leverage condition $\Lambda(\tau) \le 1$, $\Lambda = [G(K(0)) - G(K(\tau))]/\Phi_{\mathrm{ft}}(\tau)$, with G consumer fossil and Φ_{ft} the load's footprint. We advance it toward a theorem.

Theorem 3 (Leverage decomposition)**.** *The reduction is exact:* $C(\tau) - C(0) = \Phi_{\mathrm{ft}}(\tau)\,(1 - \Lambda(\tau))$, *so* $\tau = 0$ *minimizes carbon over the family iff* $\Lambda(\tau) \le 1$. *Differentiating* $C(\tau) = G(K_R(\tau), K_F(\tau)) + \Phi_{\mathrm{ft}}(\tau)$ *(with* G *independent of* K_M*),*

$$\frac{dC}{d\tau} = \frac{d\Phi_{\mathrm{ft}}}{d\tau} + (1 - F(K_F))\,K_F'(\tau) - \Big(\int_0^{K_F} F_{K_R}\Big)\,K_R'(\tau),$$

the three forces being the load's own footprint, the extra consumer fossil from a larger K_F*, and the renewable channel. The footprint force is positive and* $1 - F(K_F) \ge 0$*; signing* $dC/d\tau$ *reduces to signing the capacity responses* $K_F'(\tau)$ *and* $K_R'(\tau)$*.*

Proof. The identity is Week 3; the derivative is Leibniz, with $G_{K_F} = 1 - F(K_F)$, $G_{K_R} = -\int_0^{K_F} F_{K_R}$, and $\partial G/\partial K_M = 0$. ($\frac{d\Phi_{\mathrm{ft}}}{d\tau}$ is the total derivative, including Φ_{ft}'s dependence on the moving capacities.) □

Theorem 4 (Leverage bound under capacity monotonicity)**.** *Suppose along the budget path the fossil budget does not raise renewable capacity and does not lower fossil capacity:* $K_R'(\tau) \le 0 \le K_F'(\tau)$ *(Condition* M*). Then* $G(K(\tau)) \ge G(K(0))$*, hence*

$$C(\tau) - C(0) = \Phi_{\mathrm{ft}}(\tau) + \big[G(K(\tau)) - G(K(0))\big] \ge \Phi_{\mathrm{ft}}(\tau) \ge 0, \qquad \textit{so} \qquad \Lambda(\tau) \le 0 < 1,$$

and $\tau = 0$ *(surplus-only) is the carbon-optimal member of the family. Condition* M *holds across every shape and regime scanned—the budget raises* K_F *(it must, to serve the fossil it now draws) and does not raise* K_R *(fossil dispatch earns the* $R - c$ *margin on* fossil*, which renewables do not supply)—so surplus-only is carbon-optimal.*

Proof. $G(K_R, K_F) = \int_0^{K_F} (1 - F(x; K_R))\,dx$ has $\partial G/\partial K_F = 1 - F(K_F) \ge 0$ and $\partial G/\partial K_R = -\int_0^{K_F} F_{K_R} < 0$. Under Condition M, $K_F(\tau) \ge K_F(0)$ and $K_R(\tau) \le K_R(0)$, so G moves up on both

counts: $G(K(\tau)) \geq G(K(0))$. Substituting into the exact reduction gives the display; the inequality $\Lambda \leq 0$ is $G(K(0)) - G(K(\tau)) \leq 0$.[4] □

Remark 4 (What is now proved, and what remains). The leverage question is closed *conditionally*: $\Lambda \leq 1$ is a *theorem* given the capacity-monotonicity Condition M, and M is a transparent, checkable statement about the two capacity responses that holds throughout the scanned region. What is not proved is Condition M itself in closed form—signing $K_R'(\tau), K_F'(\tau)$ in the coupled, submodular three-capacity program requires the full inverse Hessian (Week-2's cofactor identity), which we do not carry out. So the residual open step is a clean, isolated comparative-statics sign, not the global optimality claim.

6. Numerical validation

`verify_week4.py` checks, against adaptive quadrature on Beta scale mixtures (symmetric, left- and right-skewed, mildly skewed): the classification identity $\Phi = \Delta$ to $\sim 10^{-9}$ and $\operatorname{sign}\Delta = \operatorname{sign}\Phi$ in every cell (Theorem 1); the partial signs $\partial\Phi/\partial\mu^*_{\text{low}} < 0$, $\partial\Phi/\partial\mu^\circ_{\text{mid}} > 0$ (Proposition 1); the per-mean margins $\mathrm{D}_\ell, \mathrm{D}_m$ ($\Phi_{\text{hi}} = +0.40, +0.49 > 0$) and that the exact *iff* $\mathrm{D}_\ell \Leftrightarrow$ (sweep flips) holds (Theorem 2); and Condition M (K_F non-decreasing, K_R non-increasing along τ) with the resulting $\Lambda(\tau) < 0$ (Theorem 4). The suite reports a single pass count.

[4] Verified in `verify_week4.py`: along τ, K_F is non-decreasing ($3.4 \to 5.3$) and K_R is non-increasing, so Condition M holds; $\Lambda(\tau) < 0$ throughout.